

ERIN DAVID CULLEN

SOFTWARE ENGINEER

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PERSONAL QUALITIES

- Accurate & Precise
- Detail-Oriented
- Analytically Minded
- Self-Motivated
- Always Learning

ABOUT ME

I thrive on learning and applying knowledge to build solutions that deliver value. I have a natural and unwavering drive for continuous improvement. With an understanding and involvement in mechanical and software engineering, as well as business, I can bring a blend of technical and operational insight.

In 2022, I began working on a toolset focused on improving both people and systems in the tool hire industry, applying knowledge and experience I had in business operations, logistics and systems design.

Today I work as a Software Engineer at BMW IT Hub South Africa, building enterprise-grade internal software on C# and .NET (clean architecture, Entity Framework, MassTransit) and developing AI and automation solutions with N8N and Python, while completing my BComputing dissertation.

WORK EXPERIENCE

BMW IT Hub South Africa
2026 – Present

Software Engineer

Building an enterprise-grade internal software platform on C# and .NET, following clean architecture principles: Entity Framework for data access, MassTransit for message-based communication, and Serilog for structured logging. Alongside the platform work, developing AI and automation solutions, including N8N workflows, Python tooling, and documentation automation for Jira and Confluence, with Ollama for rapid prototyping on local hardware.

C# | .NET | Entity Framework | MassTransit | Serilog | N8N

Independent Software Developer
2022 – Present

Full Stack Developer

Developing AetherFlow, a comprehensive diary system tailored for the hire industry. Collaborating with business owners and software engineers to gather domain insights and continuously refine the solution. Responsible for database architecture (PostgreSQL), backend development (Python/Flask), and frontend implementation (Next.js).

Python | PostgreSQL | Next.js | Flask | Docker

Coastal Hire Centurion
2021 – 2023

In-House IT & Systems

In-house IT and operations for the family plant and tool hire business. Full-time on site from 2021: daily operations and management (including the high-pressure hire desk), technical and mechanical work in the workshop, and requirements analysis for the Kerridge ERP implementation. Moved to part-time in 2023 when studies at Belgium Campus began, with the systems work growing into AetherFlow, a complete operations system for the business. The company network runs on a Tailscale tailnet, and technical issues are still handled remotely on the odd occasion something goes wrong.

IT Support | Networking | Tailscale | Operations | ERP

Engineering Firm (under NDA)
2021 (5 months)

Junior Engineer

Junior engineer at a mechanical engineering firm serving mining and heavy-industry clients (name withheld under NDA). Worked on part design and prototyping.

Mechanical Engineering | Part Design | Prototyping

PROJECTS

AetherFlow Diary System

Enterprise Resource Planning • Hire Industry • Full-Stack

A comprehensive diary solution designed for the hire industry. Built from real-world experience in business operations, this system addresses the unique challenges of rental, logistics and customer management.

- Diary Management
- Driver Scheduling
- Inventory Control
- Rep & CRM
- Reporting
- Database Auditing
- Logistics
- Task Management
- AI Automation
- Debtors
- Data Aggregation & Validation
- Lock Control & Rollback

Python | Flask | PostgreSQL | Next.js | TypeScript | Docker

AI Chat System

Local AI • Privacy-Focused • Conversational Intelligence

A prototype AI chat system built on Python and Ollama, designed to run as a node plugged into other software (such as AetherFlow) to automate processes like customer orders, WhatsApp communications and order status updates.

Ollama | Python | PostgreSQL | NLP

Nimbus: NLP Drone Control

Natural Language Processing • Computer Vision • Mixed Reality

An intelligent drone navigation system enabling natural language control in mixed reality environments. Voice commands like "Go to the chair" are carried out autonomously using computer vision and SLAM technology.

OpenCV | NumPy | NLTK/spaCy | PyTorch | Docker | Flask/FastAPI

More on GitHub

Academic Work • Learning Experiments

A collection of smaller projects and experiments showcasing different technologies and problem-solving approaches, with continuous learning across various programming concepts and frameworks.

Machine Learning | Jupyter | C# | Python | SQL

EDUCATION

Bachelor of Computing (Data Science)
Belgium Campus
Dissertation in progress

Bachelor of Engineering (Mechanical)
University of Pretoria
Most Innovative Project (EBIT Faculty)

National Senior Certificate
St. Albans College (IEB)

REFERENCES

Available on request

TECHNICAL SKILLS

BACKEND & DATA

C# | .NET | Entity Framework | MassTransit | Serilog | Clean Architecture | PostgreSQL | SQL Server | MongoDB | REST APIs | Microservices | Flask

AI & AUTOMATION

N8N | Python | Ollama | Agent Design | OpenAI API | Jira & Confluence Automation | Whisper | Zero-Shot Classification

FRONTEND

Next.js | React | TypeScript | Tailwind CSS

TOOLS & INFRASTRUCTURE

Docker | Docker Compose | Git & GitHub | Linux/WSL | Bash | Postman

CERTIFICATES

Mensa Member — 2024
Pretoria Chapter

Zeiss Geometric Dimensioning & Tolerancing — 2021

Most Innovative Project (EBIT Faculty, University of Pretoria) — 2019

Dale Carnegie Generation.Next — 2016

HOBBIES

- Carpentry & Woodworking
- Hiking & Nature
- Photography

RESOURCES

Website
beardedseal77.github.io/CV

GitHub
github.com/BeardedSeal77

LinkedIn
www.linkedin.com/in/de-cullen